

RESEARCH COLLABORATION | MEETING RECORD

Meeting Outcome Brief

Paper framing, P1-P2-P3 methodology, computational tests, and collaboration workflow

Meeting	Tulio-Nacho Follow-up	Date	13 August 2026
Participants	Tulio Coelho Coura; Ignacio Pérez Ortiz	Duration	27 min 6 s
Purpose	Align the introduction, methodology, computational tests, and collaboration workflow	Status	Agreed follow-up plan

Executive Summary

The meeting aligned the manuscript around a clearer technical storyline and a sharper contribution relative to prior work. The introduction is now considered structurally closer to an IEEE Transactions on Smart Grid paper, but it must more explicitly separate fairness in free-use operating envelopes from fairness-aware disaggregation of a specific service request selected by the TSO. The methodology will be organized as P1 (self-consumption), P2 (FOR computation), and P3 (fairness-aware disaggregation and validation).

A parallel computational plan was agreed. The team will quantify the speed-up obtained from transformer-by-transformer runs, test Luis's suggestion to deactivate selected constraints in the generated optimization matrix, and benchmark the complete-feeder rolling horizon. These results will determine whether the final implementation can run on the complete feeder without supercomputing resources or feeder decomposition.

CENTRAL OUTCOME Use self-consumption to establish the operating trajectory (P1), compute the grid-feasible flexibility available at the PCC (P2), and fairly disaggregate a TSO-selected feasible service point among aggregators and prosumers (P3), with empirical validation across sampled requests.

1. Agreed Paper Narrative and Research Gap

The current introduction should be retained as the main foundation. Nacho considered its literature review and overall storytelling substantially stronger than the earlier version and consistent with the style of comparable Transactions papers. The next revision should focus on differentiation, not a complete rewrite.

1.1 Distinguish Two Disaggregation Contexts

1. **Free-use flexibility:** DOE, PDOE, and P-Q DOE approaches define feasible boundaries within which users retain discretion over their operation. Fairness is commonly studied when allocating these envelopes.
2. **Requested service:** once the TSO selects a specific point from the aggregate flexibility region, that request must be converted into feasible device- or participant-level dispatch. The manuscript should emphasize fairness at this service-delivery stage.
3. **Proposed gap:** existing fairness work is concentrated on the allocation of operating envelopes or unconstrained user flexibility; comparatively less attention has been given to fairly disaggregating an activated P-Q service point while preserving distribution-network and intertemporal feasibility.

Writing implication: DOE concepts should remain as a contrast to the literature, not as an output claimed by the proposed method.

1.2 Strengthen the Multi-Period Positioning

The introduction should explain that BESS decisions in the current 30-minute interval change the state of charge and therefore the flexibility that can be offered in the next interval. The literature review must connect rolling-horizon or multi-period FOR construction with fairness and multiple aggregators, because this intersection is the most relevant point of differentiation.

4. Retain the working literature table for internal organization, but remove it from the manuscript.
5. **Build complementary must-cite lists: Tulio for FOR and multi-period flexibility; Nacho for fairness and multiple aggregators.**
6. Use approximately 35-45 references across the complete paper as a planning benchmark, subject to relevance rather than a fixed quota.
7. Review Nacho's presentation/PDF as a visual and conceptual source for revising the introduction and methodology.

2. Agreed Methodological Structure

The paper will use a three-part structure that follows the chronological decision process and gives the fairness contribution a distinct methodological role.

P1 Self-consumption baseline. Determine the operating trajectory used as the starting point of the rolling horizon. The executed BESS action fixes the state from which future flexibility is evaluated.

P2 Feasible Operation Region. Compute the network-feasible P-Q flexibility available at the PCC for the service interval, subject to distribution constraints and the BESS state inherited from P1.

P3 Fairness-aware disaggregation. Given a feasible service point requested by the TSO, allocate and dispatch the response among aggregators and prosumers while preserving feasibility and evaluating fairness.

2.1 Proposed Validation for P3

Nacho proposed testing multiple randomly or stochastically selected points from the service region. This would show whether TSO requests identified as feasible by P2 can be implemented through P3 and would provide empirical evidence on the reliability and fairness of the disaggregation. The exact sampling design, statistical metric, and confidence criterion remain to be defined.

8. Sample service requests from a clearly defined domain within the computed FOR.
9. For each request, record disaggregation feasibility, network-limit compliance, and the selected fairness metrics.
10. Report an empirical feasibility rate and uncertainty bounds; avoid committing to a value such as 99.99% until it is demonstrated.
11. Use the P3 results to verify the internal consistency of the complete P1-P2-P3 chain.

2.2 Figure Strategy

The main framework figure should be revised to show P3 explicitly, including the transition from the aggregate service point to fairness-aware dispatch. Nacho's PDF contains candidate structures for the introduction and methodology. A tentative target of three principal figures was discussed, but the final number should follow the needs of the paper.

3. Computational Plan and Performance Tests

Tulio reported that transformer-by-transformer runs reproduced the same results in less time, although the speed-up had not yet been quantified. The immediate objective is to establish whether equivalent results can be obtained more efficiently on the complete feeder.

12. Quantify the runtime reduction achieved by the transformer-by-transformer implementation before the meeting with Luis.
13. Test Luis's suggestion by deactivating the selected constraints or generated matrix rows using the GMP workflow, then compare both solutions and runtimes.
14. Run the complete-feeder rolling horizon and record solution status, total runtime, per-stage runtime, and any numerical failures.
15. Estimate the complete-day runtime under the measured percentage reduction only after equivalence has been confirmed.
16. Decide among complete-feeder execution, transformer decomposition, and supercomputing only after the benchmark results are available.

IMPORTANT QUALIFICATION The suggested 90% reduction and approximately six-hour full run were working hypotheses discussed in the meeting. They are not results and must not appear in the paper until verified.

4. Collaboration and Version-Control Workflow

The workflow will move to GitHub to avoid overwriting changes and to keep the manuscript, literature context, formulations, and supporting documents in one traceable location. Overleaf will remain the review copy shared with Luis and Mariana after the coauthors consolidate a stable version.

17. Share the current manuscript package with Nacho without replacing the more advanced working version.
18. Use commits, pushes, and pulls for manuscript edits and maintain a dedicated docs folder for literature notes, formulation notes, and AI context files.
19. Update Overleaf at review checkpoints rather than using it for simultaneous exploratory edits.
20. Use the repository instead of WhatsApp for substantive research documents and evolving technical material.

5. Decisions and Working Hypotheses

Confirmed Decisions

21. Organize the methodology as P1 self-consumption, P2 FOR computation, and P3 fairness-aware disaggregation.
22. Reframe DOE-related literature as the contrast that motivates fairness at an activated service point.
23. Revise the framework figure so that the P3 fairness/disaggregation stage is explicit.
- 24. Exchange complementary must-cite literature lists within two weeks.**
25. Use GitHub as the active coauthor workspace and Overleaf as the consolidated supervisor-review copy.

Working Hypotheses to Test

26. The fairness-at-service-point framing provides a sufficiently distinct research gap relative to DOE allocation studies.
27. Sampled P3 requests can provide meaningful empirical evidence of implementation reliability and fairness.
28. Deactivating selected constraints preserves the relevant solution while substantially reducing generation and solution time.
29. The resulting speed-up may make complete-feeder execution practical without supercomputing or transformer decomposition.

6. Action Register

No.	Action	Owner	Timing	Expected output
1	Revise the introduction to distinguish free-use envelopes from fairness-aware disaggregation of an activated service point; retain DOE literature as context.	Tulio	Next draft	Revised gap and transition paragraphs
2	Review Nacho's PDF/presentation and adapt its conceptual structure to the introduction, methodology, and framework figure.	Tulio	Immediate	Updated manuscript structure and figure brief
3	Remove the internal literature-comparison table from the manuscript while retaining it as a working research aid.	Tulio	Next draft	Clean manuscript introduction
4	Compile the must-cite FOR and multi-period flexibility papers.	Tulio	By 27 Aug 2026	Annotated FOR reference list
5	Compile the must-cite fairness and multiple-aggregator papers.	Nacho	By 27 Aug 2026	Annotated fairness reference list
6	Quantify the transformer-by-transformer runtime improvement for discussion with Luis.	Tulio	Before 14 Aug meeting	Comparable runtime summary
7	Test the model with the selected constraints active and deactivated; verify solution equivalence and measure time savings.	Tulio / Nacho	Starting 14 Aug	Controlled comparison of results and runtime
8	Run the complete-feeder rolling horizon and record total runtime, solver status, and numerical issues.	Nacho	14 Aug / weekend	Full-feeder benchmark log
9	Define the P3 sampling experiment, fairness metrics, feasibility criteria, and statistical reporting method.	Tulio / Nacho	After P3 formulation	Validation protocol
10	Create or organize the GitHub docs area and use the repository for manuscript, literature, formulation, and context files.	Tulio / Nacho	Immediate	Traceable shared workspace
11	Share the philosophical fairness reflection document when the current draft is complete.	Tulio	When ready	Additional fairness concepts for discussion
12	Update the shared Overleaf project after the coauthor version is consolidated for review by Luis and Mariana.	Tulio / Nacho	Review checkpoint	Supervisor-ready manuscript version

7. Open Questions for the Next Meeting

30. What exact sentence best states the gap between fairness in DOE allocation and fairness in activated-service disaggregation?
31. How will P3 represent fairness across aggregators and prosumers, and which metrics will be reported?
32. How should service points be sampled, and what confidence or reliability measure is appropriate?

33. Which constraints can be safely deactivated, and what evidence is needed to show that the relevant feasible set and optimal solutions are preserved?
34. Do the measured runtimes justify a complete-feeder implementation, transformer decomposition, or HPC execution?
35. Which three or four figures are essential to communicate the complete P1-P2-P3 workflow and results?

8. Proposed Agenda for the Next Checkpoint

36. Review the constraint-deactivation and complete-feeder runtime results.
37. Approve the revised introduction gap statement and merge the two literature lists.
38. Lock the P3 formulation, sampling design, and fairness metrics.
39. Review the updated P1-P2-P3 framework figure and final figure plan.
40. Confirm the GitHub-to-Overleaf review workflow and the next manuscript checkpoint.

OUTCOME STATEMENT FOR SUPERVISORS The proposed framework coordinates grid-feasible flexibility over a rolling horizon by first establishing a self-consumption trajectory, then computing the PCC Feasible Operation Region, and finally disaggregating a selected service point through a fairness-aware dispatch across aggregators and prosumers.

Source note: Prepared from the 13 August 2026 meeting transcript. Transcription errors were corrected where the intended technical meaning was clear. Unverified expectations are explicitly identified as working hypotheses.